

HR Data Analysis Capstone Project

Salt Corporation (Employee Satisfaction & Retention Analysis) - 2024

Dataset: [HR_data](#)

1. Make a copy (**File>Make a copy**)
2. Download as Microsoft Excel

Terminologies

- Attrition rate - is a key HR metric measuring the percentage of employees who leave an organization either voluntarily (resignation, retirement) or involuntarily (layoffs, termination) over a specific period without being replaced.

$$\text{Attrition rate} = (\text{Number of employees left} / \text{Total number of employees}) * 100$$

Dataset Overview

File: HR_data.csv

Records: 1,000 employees

Columns: 24 fields

Period: 2018-2020

Key Columns:

- Demographics: Age, Birth Year, Gender
- Employment: Department, Position, Employment Status, Joining Date, Years of Service
- Compensation: Salary, Benefits
- Performance: Scores (Letter & Numeric)
- Survey: Job Satisfaction (1-5), Attrition (Yes/No)
- Development: Training Hours, Last Promotion Date
- Time Off: Sick Days, Vacation Days

For more information about the columns - [Data Dictionary](#)

Project Scenario

You are hired as a data analyst at Salt Corporation in the HR department. The company surveyed 1,000 Salt Corporation employees in 2024 to measure job satisfaction and identify opportunities to keep staff motivated and engaged.

Your task: To clean the survey data, analyze patterns, and prepare a visual summary report for HR leadership to help them:

- Understand employee satisfaction levels

- Identify factors contributing to attrition
- Make data-driven retention decisions
- Optimize HR policies and programs

Deliverables

1. **Cleaned Dataset (Excel)** — Final dataset with all cleaned columns (age_clean, salary_clean, standardized categories, formatted dates), consistent data types, no unresolved missing values.
2. **Presentation Slides (PowerPoint)** — A 15-minute, 5–10 slide deck covering the project overview, data cleaning summary, descriptive analysis (salary, age, job satisfaction, department breakdowns), key insights, business interpretation, and actionable recommendations. All charts are clearly labeled and each visual answers one focused question. (*Include at least 4 answered key analysis questions*)
3. **SQL Script (Optional)** — Well-formatted, commented queries handling cleaning transformations, and aggregations. Uses uppercase keywords, CTEs, and clear aliases for full reproducibility.

Data Cleaning Step-by-Step Guide

Step 1: Initial Assessment

Check for Data quality issues per column:

- Mixed data types
- Missing Values
- Inconsistent formats e.g. dates, lettercase
- Text numbers

Confirm that the following columns have data quality issues and not down the different data quality issues identified: *age, salary, Department, Joining Date, Email, Phone Number, Employment Status, Manager ID, Sick Days Taken, Vacation Days Remaining, Last Promotion Date, Training Hours, Benefits Enrolled*

Note: Make a copy of the dataset before cleaning the data

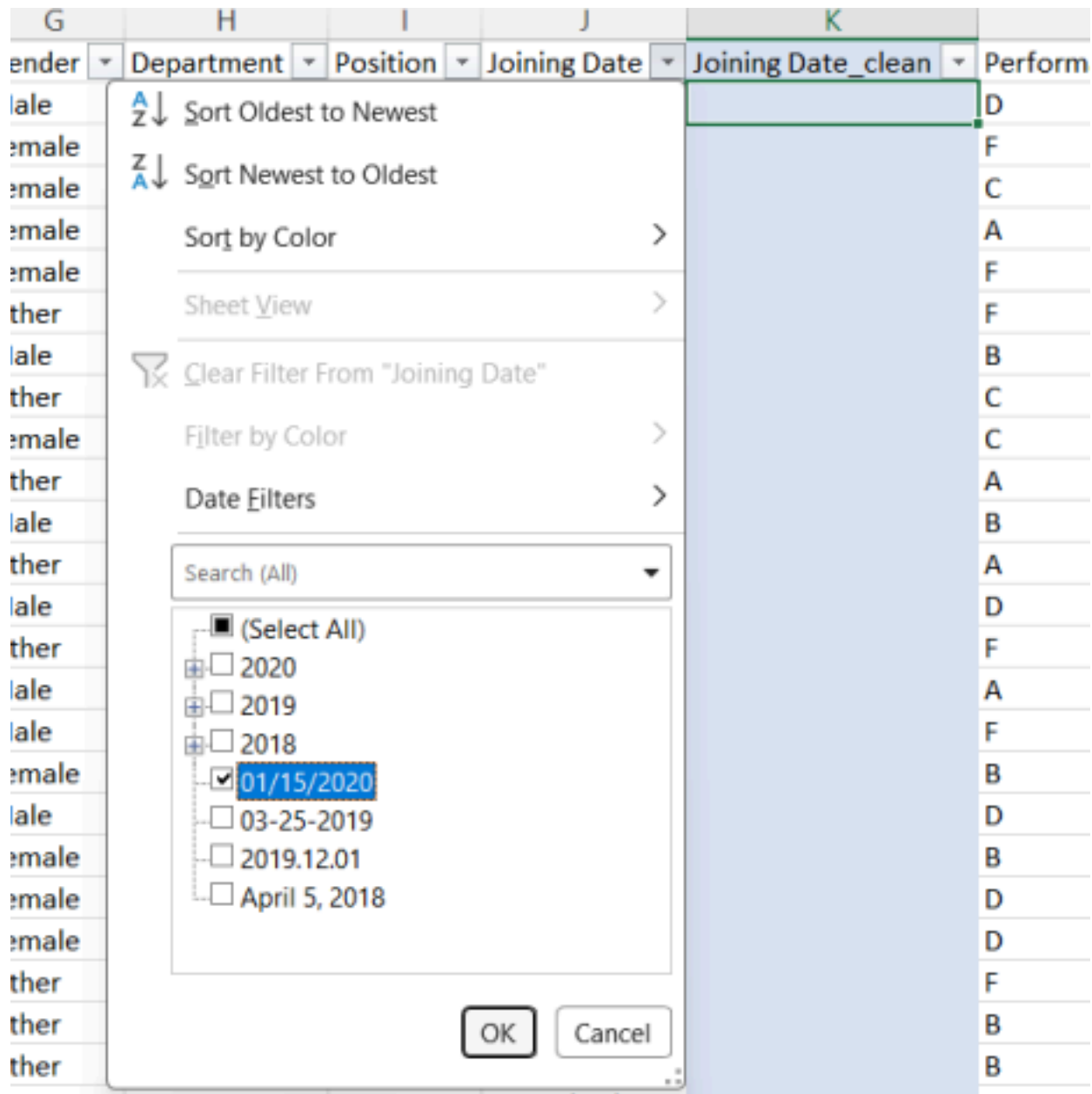
Step	Column(s)	Data Issues	Cleaning Method	New Column	Validation
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Step 2	Age	Text values (e.g., “thirty”), missing values, inconsistent formats	Create age using 2024 – Birth Year . Extract numeric birth year where needed and recalculate missing values.	age_clean	Average age = 36
Step 3	Salary	Text numbers (“SIXTY THOUSAND”), “NAN” strings, mixed formats	Convert text salaries to numeric using IF or mapping logic. Filter out “NAN”, then fill missing values using AVERAGEIFS by Department and Position.	salary_clean	Average salary = 60207.6
Step 4	Department	Inconsistent casing (finance, FINANCE, Finance, etc.)	Standardize using PROPER (optionally normalize first using UPPER).	department_clean (optional)	Values limited to Finance, HR, IT, Sales, Marketing
Step 5	Employment Status	Mixed formats (full time, FULL-TIME, contract, etc.)	Use PROPER and standardize to allowed categories.	status_clean (optional)	Only Full-Time, Part-Time, Contract
Step 6	Joining Date, Last Promotion Date	Multiple inconsistent date formats	Follow the examples from the figures after step 8.	joining_date_clean, promotion_date_clean	All rows recognized as valid date format
Step 7	Email	Missing email values	Create a new column and generate emails using CONCAT(first_name, last_name, “@saltcorp.com”) . Handle spacing/lowercase if needed using LOWER and TRIM .	email_clean	All rows contain valid company email format

Step 8	Benefits Enrolled	Inconsistent values (Yes/No variations, blanks)	Use IF logic to standardize values into Yes or No (e.g., map blanks or unexpected values appropriately).	benefits_clean	Only Yes or No values remain
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Forexample:

Create a new column as shown, then filter out one of the dates and click OK.



Input the correct date format, convert from “mm/dd/yyyy” to “dd/mm/yyyy” as shown below.

H	I	J	K	L
Department	Position	Joining Date	Joining Date_clean	Performance Score
sales	Director	01/15/2020	15/01/2020	C
Finance	Manager	01/15/2020		F
Marketing	Manager	01/15/2020		C
sales	Assistant	01/15/2020		B
	Clerk	01/15/2020		D
Marketing	Director	01/15/2020		B
sales	Clerk	01/15/2020		B
sales	Director	01/15/2020		A

Drag to autofill the remaining dates.

H	I	J	K	L
Department	Position	Joining Date	Joining Date_clean	Performance Score
sales	Director	01/15/2020	15/01/2020	C
Finance	Manager	01/15/2020		F
Marketing	Manager	01/15/2020		C
sales	Assistant	01/15/2020		B
IT	Clerk	01/15/2020		D
marketing	Director	01/15/2020		B
Sales	Clerk	01/15/2020		B
Sales	Director	01/15/2020		A
				15/01/2020

If you use google sheets, below is the date to filter.

and NAN values replaced with the department and position based average. • Department, employment status and Benefits enrolled are standardized

You may leave the missing values in the remaining columns as-is or Use or exclude the columns with a high magnitude of missing values from your analysis.

Key Analysis Questions:

1. How many employees work in each department?
2. What is the average salary in the company?
3. What is the gender distribution in the company?
4. How old are employees on average?
5. How long do employees typically stay at the company?
6. What are the job satisfaction levels?
7. How many employees left the company?
8. Which positions are most common?
9. How many sick days do employees take on average?
10. How much vacation time is unused?
11. What percentage of employees are enrolled in benefits?
12. What are the performance score distributions?
13. How much training do employees get?
14. Which department has the highest salaries?
15. Are satisfied employees less likely to leave?
16. Do high performers earn more?
17. What is the male vs female salary difference?
18. How many employees started each year?
19. Which department has the best job satisfaction?
20. Are Full-Time employees more satisfied than Part-Time?

Supplementary Exploratory Hypotheses:

1. What factors are mostly strongly associated with employee attrition? (i.e. Department Position etc)
2. How does job satisfaction vary across the organization? (e.g Department, Position, Employment Status, Age groups e.t.c)
3. Are there compensation (salary) equity issues? (e.g. Gender, Department, Position, Years of Service, Performance)

4. How effective are HR programs and development initiatives?
5. How do employment types (Full-Time, Part-Time, Contract) differ in engagement? Are part-time/contract workers less satisfied? Do they perform differently? Is turnover higher among non-full-time staff?
6. Are there differences in work-life balance across departments?
7. What are the promotion patterns and career progression opportunities? Is there a gender gap in promotions? Which departments offer the best growth opportunities?
8. How do performance ratings correlate with other HR metrics? Are high performers paid more? Are satisfied employees better performers? Do we lose our best people? Which departments have the strongest performance?

Using SQL for Data Analysis (Optional)

Watch the tutorial below to import the dataset to MySQL.

[How to import excel to mysql](#)

Remember: The goal isn't just cleaning data and making charts—it's telling a compelling story that helps Salt Corporation improve employee satisfaction and retention. Good luck with your analysis!